

BFMC 2026 – Phase 3 Status Report

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GitHub Repository: https://github.com/mahatosumit/BFMC_2026

1. Overview of Project Status

During Phase 3, Team OPTINX transformed the system into a **robust and competition-ready autonomous driving platform**, focusing on **stability, behavioral intelligence, and real-time performance**. Lane-following performance was significantly improved, achieving **reduced steering oscillations (~30–40%)** and smooth trajectory tracking. The vehicle successfully completed **5+ consecutive autonomous laps** under varying conditions.

Advanced capabilities such as **autonomous parking and highway overtaking** were implemented using a structured **Finite State Machine (FSM)** with **priority-based decision handling**, ensuring reliable transitions between driving states. The system was validated under **lighting variations, partial lane visibility, and dynamic obstacles**, demonstrating strong robustness.

2. System Architecture and Enhancements

The system follows a **distributed architecture**, where the **Raspberry Pi 5** handles perception and decision-making, and the **STM32 microcontroller** ensures deterministic control.

The perception system was enhanced with improved lane detection capable of handling **partial lane loss and lighting variations**, along with **temporal filtering** to reduce noise. A lightweight **YOLO-based pipeline** was optimized for **real-time inference (~70–80 ms)** with improved detection consistency.

The behavioral layer was structured using an **FSM**, with added states for **parking and overtaking**. A **priority-based mechanism** ensures stable decision-making in complex scenarios. The control system was refined using the **Stanley controller** and **dynamic speed adjustment**, resulting in smoother driving.

A **real-time dashboard** was used for monitoring lane detection, object detection, FSM states, and telemetry.

3. Key Innovations

- **Priority-based FSM** for deterministic decision-making
- **Temporal perception filtering** for stable detection

- **Distributed architecture (Raspberry Pi + STM32)** for reliable real-time performance
- **Real-time monitoring dashboard** for debugging and explainability

4. Task Completion Summary

All Phase 3 objectives were successfully completed. The system achieved **stable multi-lap autonomous driving** and successfully implemented **autonomous parking and overtaking**. Additional improvements included **dynamic speed control, pedestrian handling, and lane-loss recovery**.

Extensive **edge-case testing** was conducted under **lighting variation, partial lane visibility, multiple objects, and perception delays**, and issues were resolved through **iterative refinement**.

Performance optimization reduced **inference latency (~120 ms to ~75 ms)** and improved pipeline efficiency, ensuring stable real-time operation.

5. Challenges and Resolutions

- **Lane detection instability** under difficult conditions → resolved using **preprocessing and temporal filtering**
- **Latency spikes (~150 ms)** due to pipeline bottlenecks → resolved through **optimization and scheduling**
- **FSM conflicts** from multiple triggers → resolved using **priority-based decision logic**
- **Parking and overtaking inconsistencies** → improved through **iterative tuning and validation**

6. Next Steps

The system is now **competition-ready**. Next efforts will focus on achieving **zero-intervention runs**, improving **decision confidence**, and enhancing handling of **dynamic obstacles**. Additional focus will be on **final validation, robustness, and clear technical presentation** for competition evaluation.